

Phase 8G — Static Susceptibility Inference Layer

Technical Walkthrough, Location Profiling Engine & Multi-Raster Pipeline

Component: LocationProfiler (src/inference/location_profiler.py)	Model: Model A (Phase 8F Frozen)
Supported Domain: 8 Northeast Indian States (GADM 4.1)	Feature Extraction: DEM 30m, SoilGrids 250m, WorldCover 10m
Test Suite Status: 12 / 12 Tests Passed	Commit Checkpoint: 181fa54

Summary of Accomplishments

Phase 8G operationalizes the Phase 8F Model A static susceptibility model into a production-grade inference engine. Given arbitrary geographic coordinates (latitude, longitude), the engine performs official GADM boundary validation, multi-raster feature extraction (Copernicus DEM elevation, Horn's slope, aspect, relief; SoilGrids clay, sand, silt, density, taxonomy; ESA WorldCover 10m), and outputs an uncalibrated static predisposition score in [0.0, 1.0] with explainability reason codes.

Regression Samples Audit (10 Locations Across All 8 States)

Methodological Note: These samples verify pipeline execution, multi-raster extraction, and feature-to-score fidelity against known training data, not independent generalization.

Sample ID	State	Elevation	Slope	Soil Class	Land Cover	Score	Category
LS2014_000001	Arunachal	3139.7 m	37.4°	Luvisols	Tree cover	0.5848	HIGH
LS2014_000724	Meghalaya	81.7 m	15.7°	Acrisols	Tree cover	0.8779	VERY_HIGH
LS2014_001483	Manipur	289.5 m	45.7°	Acrisols	Tree cover	0.9480	VERY_HIGH
LS2014_001614	Mizoram	72.4 m	21.4°	Acrisols	Tree cover	0.8096	VERY_HIGH
LS2014_001968	Nagaland	955.5 m	33.7°	Luvisols	Tree cover	0.7329	HIGH
LS2014_001976	Sikkim	1275.4 m	39.1°	Luvisols	Tree cover	0.8218	VERY_HIGH
NEG_AR_0001	Arunachal	196.7 m	0.4°	Vertisols	Cropland	0.0073	LOW

Test Suite Verification (12 / 12 PASSED)

- Domain Enforcement:** Tawang accepted; New Delhi rejected with OUTSIDE_SUPPORTED_DOMAIN.
- Coordinate Validation:** Latitudes outside [-90, 90] and longitudes outside [-180, 180] rejected.
- Scientific Contract:** 0 rainfall and 0 proximity features passed to Model A pipeline.
- Probability Bounds:** All scores strictly bounded in [0.0, 1.0].